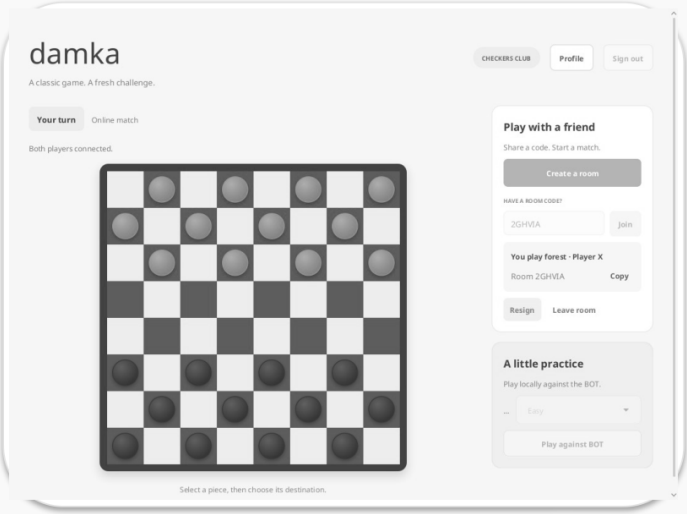
Damka

Network checkers. A shared game, a lasting profile.

JavaFX interface
TCP multiplayer · Local BOT
SQLite accounts and history

Muthanna Farhat · Rabie Abu Aad

Supervisor: Inbal Ronen



Slide 1

One game, two ways to play

Keep both screens in sync — and remember the result.

Online matches

Two computers
One room code

Local BOT

Three difficulty levels
Offline practice

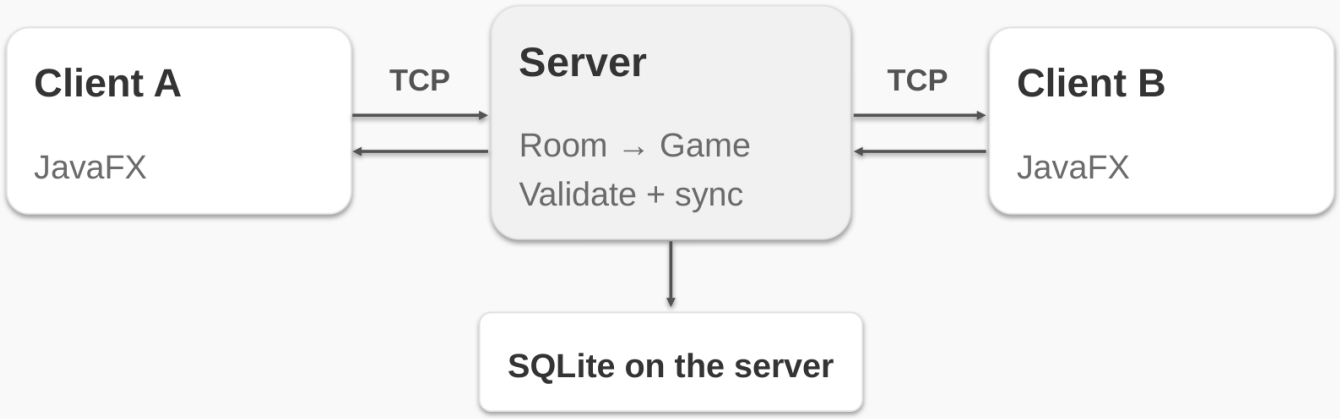
Personal profiles

Accounts and statistics
Persistent match history

One shared rules engine: Game

Slide 2

Why a client–server design?



Choice: persistent TCP for two-way updates

Alternative: HTTP polling

Slide 3

From a click to both screens



UTF-8 · one message per line · network work off the UI thread

Slide 4

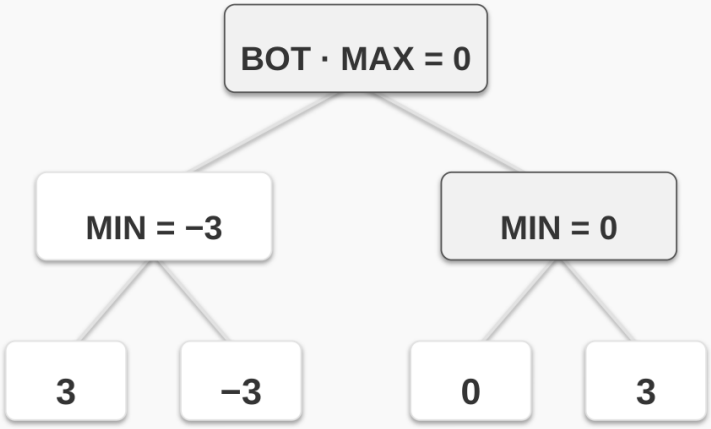
Minimax: how the BOT chooses

One rules engine: Game

Legal moves on copied states

Man = 3 King = 6

Search depth: 1 / 2 / 3

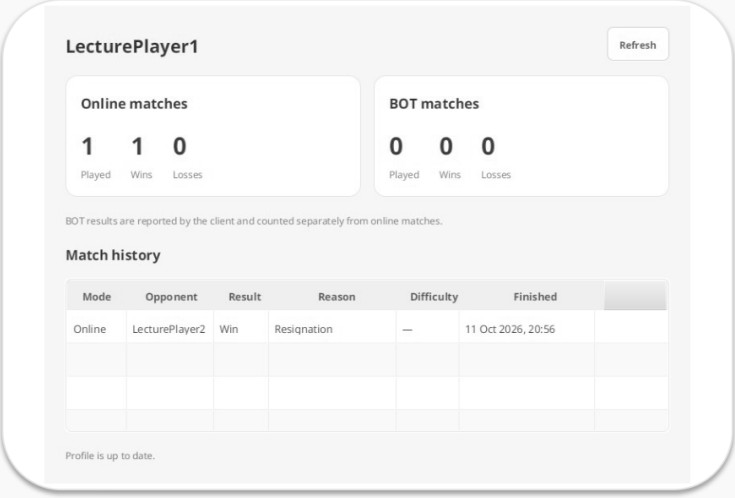


Illustrative two-step search

Alpha–Beta skips branches that cannot improve the decision.

Slide 5

Accounts that remember results



Demo account · sample history

`profiles`

Username · salt · password hash

`game_results`

Players · winner · mode · time

Statistics are derived from results.

Slide 6

Verification and AI-assisted development

48 **automated tests**

Game rules and edge cases
Real TCP protocol checks
SQLite persistence and deduplication

Documented passing run: 12 Oct 2026

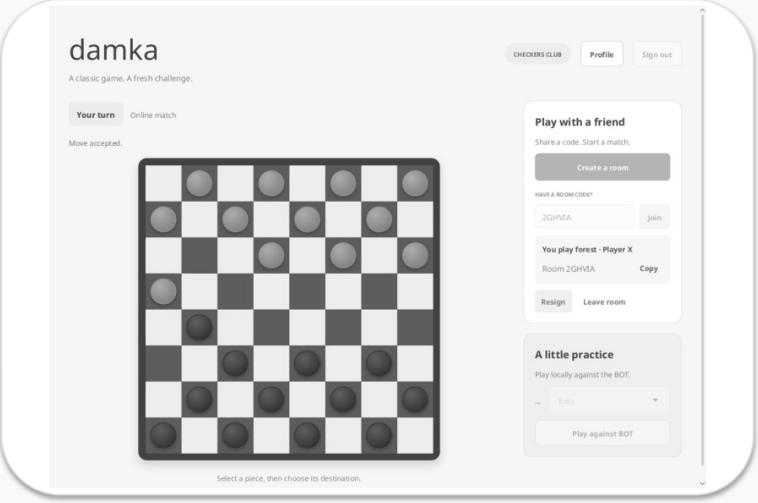
Codex + GitHub Copilot

Planning · code · explanations
Tests and documentation

Read → run → review → refine

Slide 7

Live demo — two players, one match



1. **Log in**
   Two different accounts
2. **Create / join**
   Share one room code
3. **Move / synchronize**
   Both clients show the update
4. **Resign / profile**
   One result, saved once

Slide 8

Current limits and next step

LAN focus

No TLS in this version

No match resume

A detected disconnect ends the active match

Local BOT results

Client-reported
No offline upload queue

NEXT **In-room chat over the existing TCP connection**

Slide 9

What we built and learned

One authoritative game.
Two synchronized clients.
Persistent player results.

Questions?

Game rules · Networking
BOT · Data storage

Rabie Abu Aad — Game Logic
Muthanna Farhat — TCP Communication

github.com/MuthannaFrhat/damka-game

Slide 10